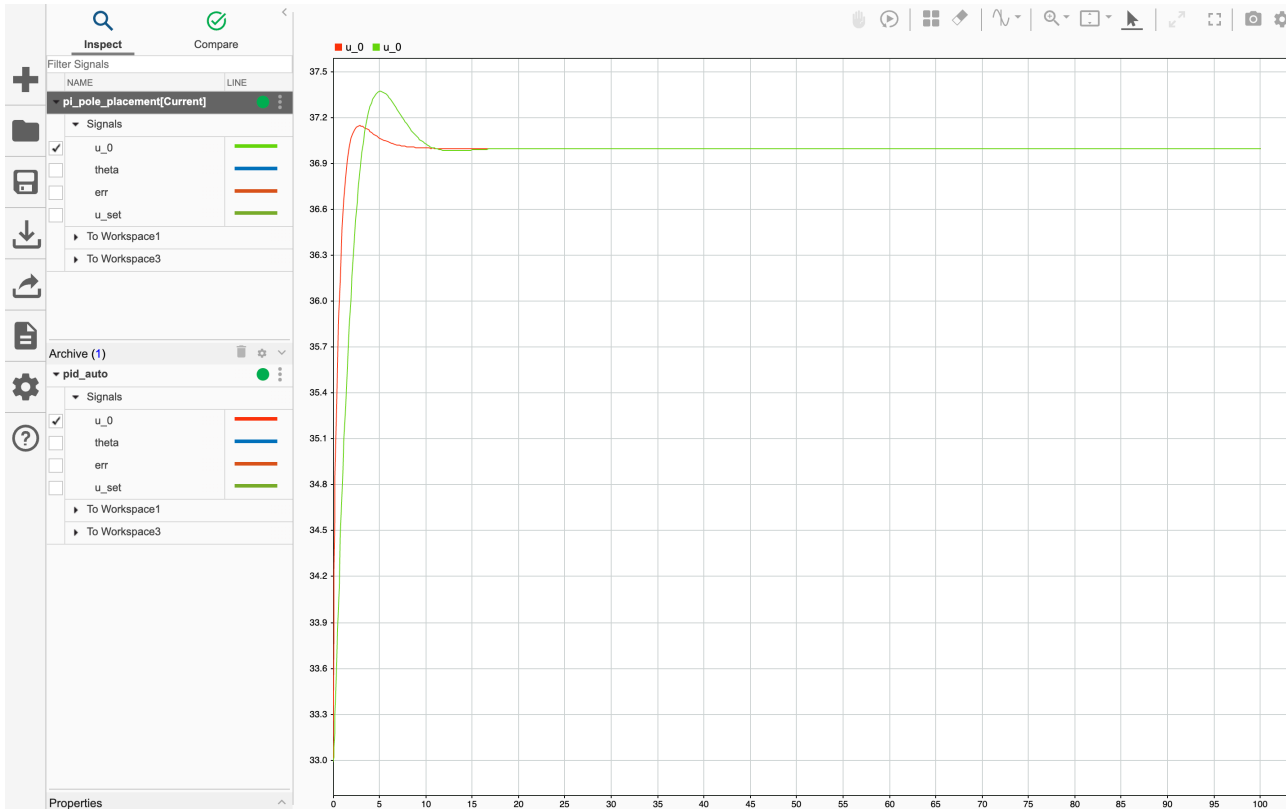


PART-I

10)

Using PID block and PID tuner app, PID parameters are set. In graph, red is response with PID controller and green is the PI controller constructed using pole placement. It can be observed that both overshoot and settling time are improved when PID and auto tuner are used.



11) Using PID controller is more advantageous compared to PI controller in following senses:

- D term allows using bigger P terms for faster response, while keeping overshoot small thanks to the braking behavior of D term.

Nevertheless, in real-world applications the inherent noise leads instability of D term, because of that generally PI controller is used.